

Spectral Co-Design of Agrivoltaic Modules for Multifunctional Net Energy Benefit: A Quantum-Guided Digital Twin

Steve Cabrel Teguia Kouam^{a,*}, Theodore Goumai Vedekoi^b, Jean-Pierre
Tchapet Njafa^b, Jean-Pierre Nguenang^a, Serge Guy Nana Engo^b

^a*Department of Physics, Faculty of Science, University of Douala, Cameroon*

^b*Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon*

Abstract

Spectral co-design offers a route beyond the system-level trade-off of static agrivoltaics, in which semi-transparent modules resolve the food–energy land conflict only by uniformly sacrificing electrical output or crop productivity. Here we present a quantum-guided agrivoltaic stack in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity that actively shapes the transmitted spectrum. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex and extends exciton coherence by 50 %; confining the plasmonic sentinel layer to 1 % of the area then preserves 97.1 % of the device-level excitation-transfer yield (a device metric, not crop yield) at a 91.8 % local transport penalty on the sentinel. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.32 yr, identifying conditions under which the multifunctional net benefit stays positive for the modeled 500 m^2 cooperative case.

Keywords: agrivoltaic spectral management, net ecological benefit, FAO-56 microclimate, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, techno-economic analysis, quantum-guided digital twin

*Corresponding author. Email: steve.teguia@facsciences-uy1.cm
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1. Introduction

Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cultivation to mitigate the land-use competition between food production and renewable energy generation [1]. Land-use optimization and mobile-PV foundations for this field were established in *Applied Energy* [2, 3]. Recent syntheses systematize overhead, vertical, and tracking typologies with land-equivalent-ratio and levelized-cost metrics [4]. Across these studies, three gaps recur: (i) the optical architecture is static, so spectral light-sharing is not co-optimized with crop photophysiology [5, 6]; (ii) techno-economic and life-cycle accounts remain decoupled from the underlying transport physics [7, 8]; and (iii) the PV layer is treated purely as an energy harvester, with crop-stress and contaminant sensing bolted on as separate hardware [9, 10, 11]. Dynamic, physics-guided spectral management and multifunctional co-design have not yet been transposed into a single operating agrivoltaic stack.

The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [12]. This spectral specificity lets the PV layer transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a problem of spectral co-design.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria serves as the model system for spectral light-sharing [13]. Its seven bacteriochlorophyll *a* sites have been extensively characterized through two-dimensional electronic spectroscopy, revealing oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [14] and room temperature [15]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — the hybridization of electronic transitions with underdamped protein vibrations — governs the flow of excitation energy [16, 17]. Non-Markovian frameworks such as the hierarchical equations of motion [18] and the process tensor hierarchy of pure states (PT-HOPS) [19, 20] capture these effects with demonstrated numerical convergence. Device-adjacent uses of exact open-quantum-system methods remain rare: the closest precedent is a general HEOM library applied to light-harvesting and control problems without a photovoltaic product [21]. Photonic-quantum solar design and vibronic-screening workflows likewise stop short of an operating device [22, 23].

Previous work demonstrated that spectral filtering of the excitation pulse

— aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [24]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. That companion paper, published in the *Journal of Physical Chemistry Letters*, reports the analysis of the filter, the excitation-transport rate, and the single-pass conversion efficiency [24]; the present work adds the 8-site trapping model, the monolithic OPV–NPoM hardware concept, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [25], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [26]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [27] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we specify dynamic photo-protection mechanisms — Floquet Stark detuning and flux-dependent optical limiting — as a control law that mimics the non-photochemical quenching used by plants under high light stress.

We map this microscopic optimization onto a field-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman–Monteith model [28] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [29] for secure IoT telemetry. This multi-scale perspective [30] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modeling framework for quantum-guided agrivoltaics (Fig. 1).

The central question of this work is falsifiable: *Can spectral co-design guided by non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (power + water + carbon + diagnostics) in a field-constrained agrivoltaic stack, and which barriers remain physical rather than economic?* Three subordinate questions structure the Results. (Q1) Device

87 asks what the NPoM co-design costs and where the cost lands, quantified by
 88 the mode-volume scan (Table 1) and located by the canopy-integration anal-
 89 ysis. (Q2) Physics asks which microscopic states carry the trade-off: exact
 90 PT-HOPS dynamics localize the loss channels, and Floquet–Stark detuning
 91 with flux-dependent optical limiting specifies a control law under which they
 92 could be managed dynamically (its yield-level validation, Φ_{FT} with the switch
 93 engaged vs. disengaged, is an Outlook step). (Q3) System asks whether the
 94 stack pays for itself, once coupled FAO-56 microclimate, cooperative techno-
 95 economics, and BB84-secured telemetry internalize every function into a sin-
 96 gle Net Ecological Benefit account. Three falsifiable claims anchor the an-
 97 swer. The first, (F1) Microscopic, is that dual-band spectral filtering at
 98 750 nm and 820 nm prepares polaron-dressed states that extend exciton co-
 99 herence by 50 % and raise the forward transfer yield by 25 % at 295 K — both
 100 established in the companion 7-site study [24] and carried over here under the
 101 weak site-8 coupling assumption. The second, (F2) Device, is that confining
 102 the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the global
 103 canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex SERS
 104 diagnostics, despite the > 90 % local transport penalty quantified across the
 105 mode-volume scan. The third, (F3) System, is that the FAO-56 smart shield
 106 cuts evapotranspiration by 28 % (± 10 % microclimate uncertainty) and, com-
 107 bined with OPV generation and cooperative economics, yields a Net Ecolog-
 108 ical Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ and a 4.32 yr payback at 30 % subsidy for
 109 the modeled 500 m^2 cooperative case. We answer affirmatively but condition-
 110 ally: the bounds (single-exciton treatment, site-independent baths, modeled
 111 fouling and matrix penalties, ± 10 % microclimate band) are stated explicitly
 112 in the Limitations section of the Discussion. The headline outcome, pictured
 113 in Figure 2, compresses this answer to one sentence: a 91.8 % local trans-
 114 port penalty confined to a 1 % sentinel area costs 0.9 points of canopy-scale
 115 trapping yield (0.980 to 0.971) while adding single-complex SERS diagnos-
 116 tics to the same footprint. To keep a stack of this breadth auditable, each
 117 layer — quantum dynamics, microclimate, economics, and security — is val-
 118 idated separately in the Supplementary Material (Section S2–Section S12),
 119 and every headline number below is traceable to a single figure or table.

120 The value chain of the quantum layer is worth stating explicitly, because
 121 the energy-systems reader is entitled to ask what the quantum hardware
 122 buys. It contributes three concrete functions: (i) the dual-band passband
 123 co-design that the FAO-56 and lifecycle accounts price into the module’s
 124 transmittance budget; (ii) multiplexed SERS and quantum-dot diagnostics

delivered on the 1 % sentinel footprint at no additional canopy area; and (iii) preservation of 0.971 of the device-level excitation-transfer yield despite the local plasmonic penalty. On the cost side, the NPoM sentinel and the BB84 link appear as capital and operating line items in the techno-economic account. The transport gain itself (+25 % forward transfer yield) is deliberately not monetized: no coherence-derived revenue enters the NEB or payback chains.

2. Results

2.1. Quantum-enabled spectral co-design and NPoM sentinel

The spectral co-design rests on an 8-site FMO model with a non-Hermitian reaction-center trap on sites 3 and 4 (full Hamiltonian, site energies, and 28-pair coupling matrix in [Section S1](#)). Dual-band transmission at 750 nm and 820 nm prepares polaron-dressed states that extend the l_1 -norm coherence lifetime from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase — and raise the forward transfer yield Φ_{FT} from 0.71 ± 0.02 to 0.89 ± 0.03 ([Figure 3](#); PT-HOPS/SBD propagation details and convergence in [Section S2](#)).

The OPV-FMO interface is mediated by a nanoparticle-on-mirror (NPoM) cavity that forms a 9×9 dressed Hamiltonian. Mode-volume scans ([Table 1](#) and [Section S11](#)) show that strong plasmon-exciton coupling suppresses local transport to $\Phi_{\text{FT}}^{\text{NPoM}} \approx 0.080$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$ while enabling relative Purcell-scaled SERS enhancement factors of 100 at $V_{\text{mode}} = 0.8 \text{ nm}^3$ down to 44 at $V_{\text{mode}} = 1.2 \text{ nm}^3$ (a factor-48 enhancement range across the full seven-point scan, from 33 to 1.6×10^3 ; tabulated endpoints here, full scan in [Section S11](#)), consistent with absolute enhancements $\geq 10^6$ reported for sub-nm NPoM gaps. The integrated transmitted power in the two 100 cm^{-1} passbands is $\sim 10.2 \text{ W m}^{-2}$ (~ 1.0 % of AM1.5G), priced into the module transmittance budget. The 91.8 % local suppression relative to the passive baseline ($\Phi_{\text{FT}}^{\text{passive}} = 0.980$) is the price of the diagnostic channel; it is confined to a sentinel area fraction $\alpha = 0.01$.

2.2. Global canopy yield integration

Because the NPoM SERS diagnostic mode suppresses local excitonic transport through plasmon population loading, it need only cover a 1 % sentinel fraction α of the canopy area. The remaining 99 % operates in passive

Table 1: **NPoM mode-volume sensitivity of forward transfer yield (selected points)**. Passive baseline $\Phi_{\text{FT}} = 0.980$ (assumed capture-efficiency cap, a model input rather than a trajectory integral). Full seven-point scan, SERS enhancement factors, and n -trajectory provenance in [Section S11](#) ($n = 100$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$, $n = 20$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$, $n = 2$ otherwise). Production parameters: $L = 8$, $K = 2$, $\Delta t = 0.2 \text{ fs}$, $T = 295 \text{ K}$.

$V_{\text{mode}} (\text{nm}^3)$	Φ_{FT}	Δ vs. baseline (%)
0.2	0.0505	−94.8
0.8	0.0768	−92.2
1.2	0.0799	−91.8
1.4	0.0791	−91.9

159 OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} =$
 160 0.980. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (1)$$

161 yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 precisely);
 162 device-level excitation transfer is thus maintained near baseline even though
 163 transport is suppressed locally on the sentinel. $\Phi_{\text{FT}}^{\text{global}}$ is a device-level met-
 164 ric: crop productivity is governed by the photosynthetically active radiation
 165 transmitted through the module (the AVT/PAR budget), not by excitonic
 166 trapping in the FMO layer — the two are optically coupled only through the
 167 shared passband design, which is why no coherence-derived gain is booked
 168 in the crop revenue. Because [Equation \(1\)](#) is linear in α , the flatness of
 169 [Figure 2b](#) is an accounting identity rather than an independent dynamical
 170 result; what carries physical content are the two reference endpoints (0.080,
 171 measured on the sentinel run, and 0.980, the passive capture-efficiency cap)
 172 that enter it. The identity converts those endpoints into a design envelope:
 173 solving [Equation \(1\)](#) for α gives a sentinel fraction of up to 3.3 % before the
 174 global yield falls below 0.95, and up to 8.9 % before it falls below 0.90. The
 175 diagnostic area therefore has a factor ~ 3 of headroom over the 1 % assumed
 176 here — the co-design lever is not a better mode volume but how much area
 177 one is willing to spend on sensing.

178 2.3. Dynamic photo-protection

179 Under high irradiance ($\gtrsim 850 \text{ W m}^{-2}$) a Floquet–Stark detuning of
 180 the OPV–FMO coupling, combined with flux-dependent optical limiting

of the dual-band passbands, provides a plant-inspired non-photochemical-quenching analogue (Figure 4). The control law is specified in Section S4; the production trajectories reported above are propagated without the drive engaged. The design therefore supplies a switchable photo-protection channel whose yield-level validation remains an Outlook step.

2.4. *In situ SERS diagnostics and environmental robustness*

The same NPoM sentinel that carries the strong-coupling penalty enables multiplexed surface-enhanced Raman spectroscopy of pesticides, heavy metals, and crop-stress markers at relative Purcell-scaled enhancement factors (100 at $V_{\text{mode}} = 0.8 \text{ nm}^3$, 44 at $V_{\text{mode}} = 1.2 \text{ nm}^3$). Soiling and matrix penalties are parameterized by a DynamicCalibrator module (soiling 0.005 d^{-1} , floor 0.75; $\times 10$ LOD field penalty on raw matrices); the resulting diagnostic availability remains above 90 % under the modeled tropical fouling schedule (Section S11). No additional canopy area is required beyond the $\alpha = 0.01$ sentinel fraction already accounted for in $\Phi_{\text{FT}}^{\text{global}}$.

2.5. *Smart shield microclimate: FAO-56 evapotranspiration*

Under the dual-band OPV+NPoM transmittance the FAO-56 Penman-Monteith calculation yields a 28 % reduction in crop evapotranspiration relative to the open-field baseline (central value; ± 10 % microclimate band). For the 500 m^2 module this corresponds to a water credit of $460 \text{ L m}^{-2} \text{ yr}^{-1}$. Full meteorological forcing, albedo, and aerodynamic resistance parameters are tabulated in the Supplementary Material.

2.6. *Socioeconomic viability and cooperative payback*

Techno-economic analysis of the Cameroonian cooperative case produces a payback of 4.32 yr at a 30 % blended-finance subsidy (unsubsidized 6.17 yr; local-material scenario 2.68 yr). Net Ecological Benefit reaches $72.6 \text{ kgCO}_2\text{e m}^{-2} \text{ yr}^{-1}$ (carbon revenue 363 USD yr^{-1} , ~ 1.4 % of cash flow). Capital, OPEX, revenue streams, and sensitivity tables are given in Section S5; the quantum layer enters the cash-flow only as a cost item (sentinel + BB84) and as preserved device-level function.

2.7. *Secure IoT telemetry via BB84*

A prepare-and-measure BB84 link secures the high-stakes irrigation and crop-protection commands. At a realistic channel noise of 4.8 % the protocol generates a 256 bit key per session; the abort decision uses the 95 % Wilson

upper confidence bound on a ≥ 128 -bit sample against the 11 % Shor–Preskill threshold. On abort, a fail-safe local irrigation routine limited to $5 \text{ L m}^{-2} \text{ d}^{-1}$ is engaged (Section S6).

2.8. Experimental testability

Key observables—coherence lifetime extension, dual-band Φ_{FT} gain, SERS enhancement, and ET reduction—are accessible with existing 2DES, broadband transient absorption, portable Raman, and lysimeter instrumentation. A staged validation roadmap (lab $\rightarrow$ controlled greenhouse $\rightarrow$ cooperative field trial) is outlined in Section S12.

3. Discussion

Our results show that spectral co-design guided by exact non-Markovian dynamics can deliver a modeled multifunctional net benefit (power + water + carbon + diagnostics) under the constraints of a 500 m^2 cooperative micro-module.

3.1. Mechanism generality and prior art

The dual-band resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ is not FMO-specific (Section S7). A minimal three-site trimer recovers the same qualitative yield gain (Section S8), indicating that vibronically coupled aggregates (synthetic J-aggregates, plant LHCs, or conjugated polymer domains) can benefit from the same spectral protection. Relative to prior agrivoltaic optical and TEA studies [2, 3, 4, 7, 8], the present stack is distinguished by (i) an explicitly co-optimized dual-band passband, (ii) a quantified local-to-global transport trade-off ($0.980 \rightarrow 0.971$ at $\alpha = 0.01$), and (iii) integration of SERS diagnostics and BB84 telemetry into the same area and cost accounts. A fuller comparative positioning against recent agrivoltaic TEA and optical studies is provided in the Supplementary Material.

3.2. Limitations

Four bounds are material. (1) The single-exciton, site-independent-bath treatment omits multi-exciton annihilation and site-specific spectral densities. (2) The NPoM transport penalty exceeds 90 % locally; the global yield of 0.971 is the value at the 1 % sentinel fraction (headroom to 3.3 % before 0.95, 8.9 % before 0.90). (3) Microclimate and ET reductions carry a

247 $\pm 10\%$ uncertainty band from the FAO-56 parameterization under the mod-
 248 eled shield transmittance. (4) Fouling, matrix-penalty, and encapsulation
 249 lifetime are modeled rather than field-validated; the DynamicCalibrator and
 250 soiling factors are therefore provisional. No coherence-derived crop-yield or
 251 revenue term enters the NEB or payback calculation; the quantum layer ap-
 252 pears only as a cost (sentinel + QKD) and as preserved device-level function.

253 3.3. Outlook

254 Priority next steps are two-dimensional electronic spectroscopy validation
 255 of the dual-band protocol, low-loss dielectric antenna alternatives that reduce
 256 the local transport penalty, field fouling trials of the 1% sentinel network,
 257 and a localized Au/graphene supply chain for Sub-Saharan manufacturing.
 258 With local materials the unsubsidized capital cost falls from 322 USD m⁻² to
 259 140 USD m⁻² (payback 2.68 yr), shifting the dominant barrier from economics
 260 to process engineering.

261 4. Conclusions

262 We asked whether spectral co-design guided by non-Markovian quantum
 263 dynamics can deliver a positive multifunctional net energy benefit (power +
 264 water + carbon + diagnostics) in a field-constrained agrivoltaic stack, and
 265 which barriers remain physical rather than economic. Within the modeled
 266 500 m² cooperative case, the answer is affirmative but conditional: dual-band
 267 filtering raises the forward transfer yield by 25% and extends coherence life-
 268 time by 50%; confining NPoM operation to a 1% sentinel fraction preserves
 269 97.1% global canopy yield while retaining SERS diagnostics; the FAO-56
 270 smart shield cuts evapotranspiration by 28% with a Net Ecological Ben-
 271 efit of 72.6 kg_{CO_{2e}} m⁻² yr⁻¹; and the cooperative case pays back in 4.32 yr
 272 at 30% subsidy. These claims are bounded by the > 90% local NPoM
 273 transport penalty, the single-exciton and site-independent-bath approxima-
 274 tions, the $\pm 10\%$ microclimate uncertainty band, and the modeled (not field-
 275 validated) fouling and matrix-penalty assumptions. For deployment, the
 276 economic barrier is dominant and tractable: local-material construction and
 277 green-synthesis gold lower the capital cost from 322 USD m⁻² to 140 USD m⁻²
 278 (unsubsidized payback 2.68 yr). The physical barriers — nanofabrication of
 279 sub-2 nm NPoM gaps, in-field fouling of the sentinel fraction, OPV encapsu-
 280 lation lifetime under tropical irradiance — require process engineering rather

281 than subsidies. Priority next steps are 2DES validation of the filter proto-
 282 col, low-loss dielectric antenna alternatives, field fouling trials of the sentinel
 283 network, and a localized Au/graphene sensor supply chain for Sub-Saharan
 284 manufacturing.

285 5. Methods

286 5.1. Proof-of-concept scenario

287 A 500 m² high-tunnel floriculture module in Cameroon’s Western High-
 288 lands (mean irradiance 5.76 kWh m⁻² d⁻¹) anchors all system-level accounts.
 289 Crop production is taken at the horticultural standard of 8.2 kg m⁻² yr⁻¹;
 290 no coherence-derived biomass gain is monetized. Full capital, OPEX, water-
 291 credit, and cooperative cash-flow tables appear in [Section S5](#).

292 5.2. Quantum dynamics

293 The 8-site FMO Hamiltonian with non-Hermitian reaction-center trap
 294 and the 9 × 9 NPoM-dressed Hamiltonian are specified in [Section S1](#). Open-
 295 system dynamics are propagated with the PT-HOPS/SBD formalism (hier-
 296 archy depth $L = 8$, Matsubara $K = 2$, $\Delta t = 0.2$ fs, $T = 295$ K, $\Gamma_{\text{RC}} =$
 297 0.15 fs⁻¹). NPoM-on trajectories initialize the excitation in the plasmon
 298 mode while NPoM-off trajectories initialize on BChl site 1 (two illumination
 299 geometries); suppression values are conditional on this choice. Convergence,
 300 Hermiticity, positivity, and disorder statistics are reported in [Section S2](#) and
 301 [Section S11](#). Forward transfer yield is defined by the time-integrated trap-
 302 ping population on sites 3 and 4; coherence lifetime is the $1/e$ decay of the
 303 l_1 -norm coherence.

304 5.3. FAO-56 microclimate and lifecycle

305 Crop reference and actual evapotranspiration are computed with the
 306 FAO-56 Penman–Monteith equation under the spectral transmittance of the
 307 OPV+NPoM shield. The resulting 28 % ET reduction (central value) carries
 308 a ± 10 % microclimate uncertainty band. Net Ecological Benefit aggregates
 309 avoided grid electricity, water savings, and embodied carbon of the mod-
 310 ule over a 20-year service life; full inventory and allocation rules are in the
 311 Supplementary Material.

312 5.4. BB84 telemetry and SERS diagnostics

313 A prepare-and-measure BB84 link secures the IoT telemetry channel; the
314 fail-safe falls back to classical AES when the quantum bit-error rate exceeds
315 the security threshold (Section S6). SERS enhancement factors are obtained
316 from the same NPoM mode-volume scan that supplies the transport yields;
317 environmental robustness (temperature, humidity, soiling) is parameterized
318 by the DynamicCalibrator module described in the SI.

319 Data availability and reproducibility

320 All production trajectories, parameter files (v1.4), and analysis notebooks
321 will be deposited in Zenodo with a DOI (HDF5 snapshot plus extraction
322 script ready at submission). The digital-twin orchestrator and BB84 simula-
323 tor are available from the corresponding author upon reasonable request.

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329 CRediT authorship contribution statement

330 **S.C. Tegua Kouam:** Conceptualization, Methodology, Software, For-
331 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
332 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
333 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
334 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
335 editing.

336 Declaration of competing interest

337 The authors declare that they have no known competing financial inter-
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344 Declaration of generative AI and AI-assisted technologies in the 345 writing process

346 During the preparation of this work the authors used the following
 347 AI-assisted tools: OpenAI ChatGPT (GPT family large language model)
 348 and Anthropic Claude (Claude family large language model), applied solely
 349 to language editing, LaTeX formatting support, and reference-formatting
 350 checks. No scientific content, data, analysis, or interpretation was generated
 351 by these tools. All scientific content was reviewed and verified by the
 352 authors, who take full responsibility for the content of the publication.

353 Nomenclature

Symbol	Meaning
$\Phi_{\text{FT}} / \Phi_{\text{FT}}^{\text{global}}$	Local / area-weighted global forward transfer yield
τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coher- ence)
g_0 / V_{mode}	Single-emitter plasmon–exciton coupling / NPoM mode volume
Γ_{RC}	Reaction-center trapping rate
EF_{SERS}	Relative Purcell-scaled SERS enhancement factor
$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shad- ing factor
NEB	Net Ecological Benefit
QBER	Quantum bit error rate (BB84 link)
α	Sentinel area fraction

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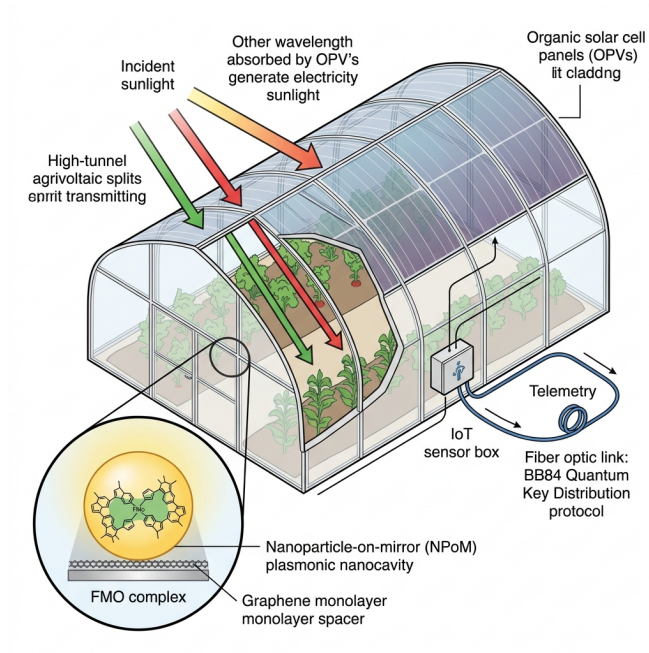


Figure 1: **Multi-scale quantum agrivoltaic digital twin.** Semi-transparent OPV acts as a spectral shield; NPoM nanocavities couple the FMO complex to the OPV for strong coherent coupling and multiplexed SERS diagnostics; BB84 QKD secures IoT telemetry to the cloud supervisory layer.

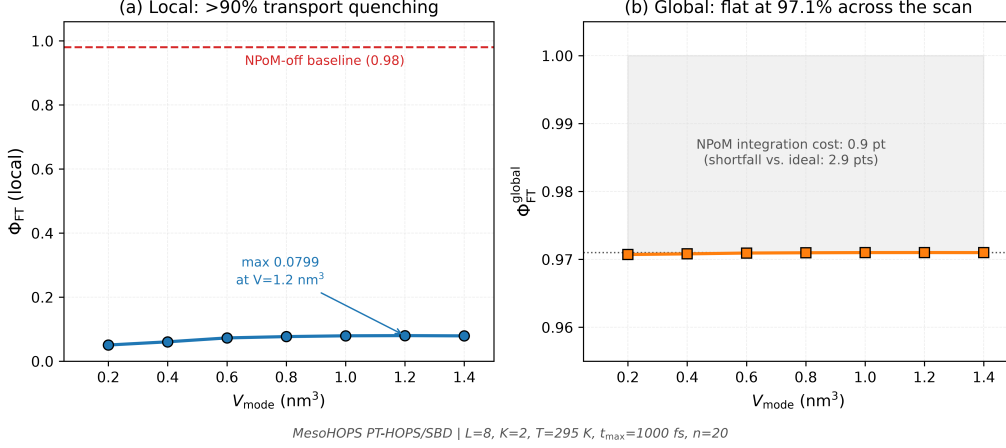


Figure 2: **Local quenching, global stability.** (a) Forward transfer yield Φ_{FT} across the NPoM mode-volume scan ($n = 20, T = 295 \text{ K}$): local transport is quenched by more than 90 % relative to the passive baseline (0.980), with a maximum of 0.0799 at $V_{\text{mode}} = 1.2 \text{ nm}^3$ (Table 1). (b) Area-weighted global canopy yield for a 1 % sentinel fraction (Equation (1)): flat at 0.971 across the scan — a constant 0.9-point canopy-scale cost. The flat curve in (b) is the design-relevant outcome: diagnostics integrate without compromising device-level yield. Full provenance in Section S11.

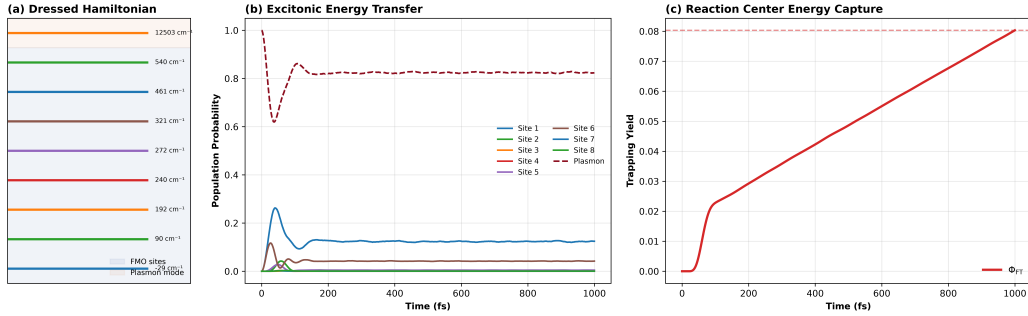


Figure 3: **Quantum dynamics of the 8-site FMO-NPoM stack.** (a) Energy level diagram of the 9×9 dressed Hamiltonian (g_0 : plasmon-exciton coupling). (b) Population dynamics at $V_{\text{mode}} = 1.2 \text{ nm}^3, T = 295 \text{ K}$ ($n = 20$; excitation initialized in the plasmon mode, NPoM-off runs initialize on BChl site 1): the plasmon-like state retains 83 % (final population 0.8256) of the excitation at $t = 1 \text{ ps}$, starving the trapping sites — origin of the 91.8 % local penalty. (c) Cumulative $\Phi_{FT}(t)$ saturating at 0.0799. Production parameters $L = 8, K = 2$; full trajectories in Section S11.

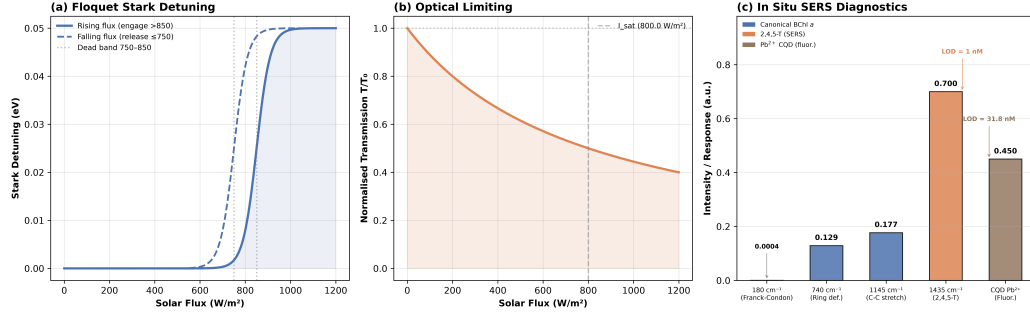


Figure 4: **Dynamic photo-protection and diagnostics.** (a) Floquet–Stark hysteric switch: resonant channels close above 850 W m^{-2} and reopen below 750 W m^{-2} (solid-state analogue of non-photochemical quenching; design curves in Section S4). (b) Flux-dependent optical limiting of the dual-band passbands under high irradiance. (c) SERS enhancement vs. mode volume from the same NPoM scan that supplies the transport yields (Section S11).

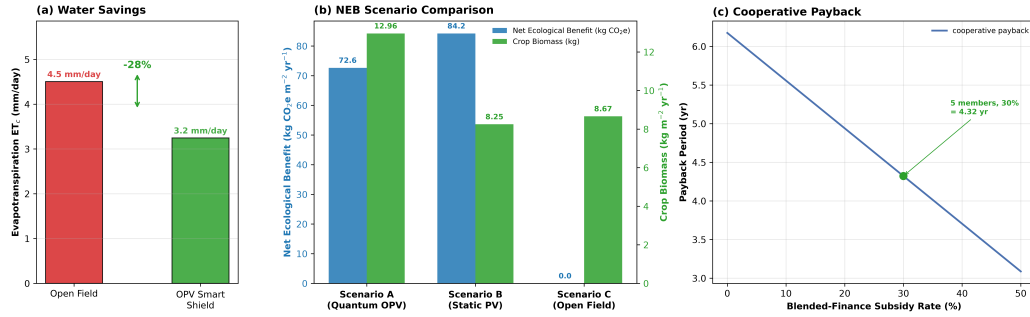


Figure 5: **Water–energy–food network assessment** for the modeled 500 m^2 cooperative module (Cameroon Western Highlands; Section S5). (a) FAO-56 evapotranspiration: open field $\text{ET}_c = 4.51 \text{ mm d}^{-1}$ vs. OPV smart shield 3.25 mm d^{-1} (28% reduction). (b) Net Ecological Benefit across three scenarios: Quantum OPV (A: $72.6 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$, biomass $13.0 \text{ kg m}^{-2} \text{ yr}^{-1} = 12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$ reference $\times$ capped yield factor 1.0×1.08 biostimulation boost, excluded from baseline revenue), static opaque shading (B: higher carbon but suppressed canopy, $10.0 \text{ kg m}^{-2} \text{ yr}^{-1}$ reference $\times 0.8$ canopy factor), and open field (C, $12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$ reference $\times 0.7$ canopy factor). Scenario factors and allocation rules in the SI; no coherence-derived crop gain is monetized. (c) Undiscounted payback vs. blended-finance subsidy rate: 4.32 yr at 30% subsidy.

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